

ELEC 221 Lecture 02

LTI systems, DT impulse response and the convolution sum

Tuesday 15 September 2026

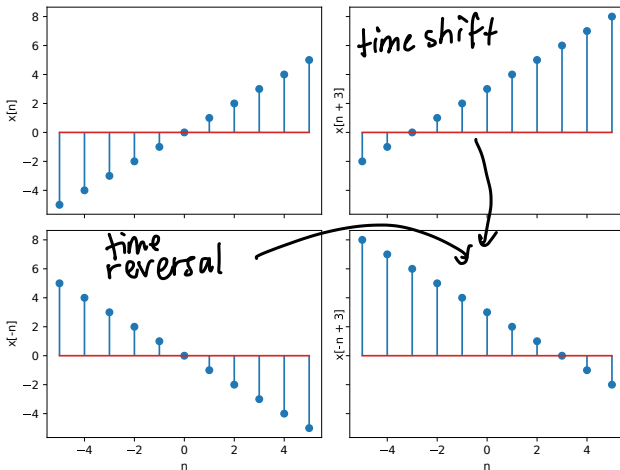
Announcements

- Quiz 1 today
- Thursday class also on Zoom

Last time

We applied simple transformations to continuous-time and discrete-time signals.

$$x(t) \quad x[n]$$



Last time

Systems respond to signals and output new (transformed) signals.

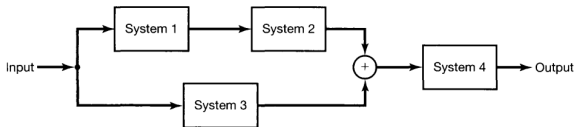
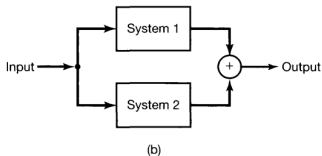
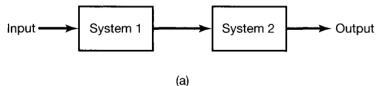


Image credits: Signals and Systems 2nd ed., Oppenheim

Last time

We explored some properties of systems:

- ① Memory
- ② Invertibility
- ③ Causality
- ④ Stability
- ⑤ Linearity
- ⑥ Time invariance

Quiz time

Available at us.prairielearn.com

- Work individually
- You may use lecture slides, notes, or the textbook.
- Please do not search online or use AI for answers

Once you start the quiz you will have 10 minutes to complete it.

You have multiple attempts per question (discounted score).

Questions will re-enabled later for practice with random variants.

Today

Learning outcomes:

- Define what it means for a system to be LTI (linear, time-invariant)
- Define the DT unit impulse and unit step functions
- Define the convolution sum and use it to compute the output of a system

Linearity

★ horizontal line test
(look up)

Consider a function f such that $y = f(x)$.

tests injectivity
(one-to-one -ness)

If f is linear, what key properties does it have?

$$f(ax) = a \cdot f(x)$$

homogeneity

$$f(x_1 + x_2) = f(x_1) + f(x_2)$$

additivity

$$f(x) = \underline{x+1}$$

$$f(ax) = ax + 1 \neq a \cdot f(x)$$

linear as in
line
but not linear
as in function
linear

Properties of systems: linearity

A **linear** system $x(t) \rightarrow y(t)$ sends $x(t) \rightarrow \boxed{S} \rightarrow y(t)$

$$x'(t) = a x(t) \rightarrow \boxed{S} \rightarrow y'(t) = a \cdot y(t)$$

$$x_1(t) \rightarrow \boxed{S} \rightarrow y_1(t)$$

$$x_2(t) \rightarrow \boxed{S} \rightarrow y_2(t)$$

$$x_1(t) + x_2(t) \rightarrow \boxed{S} \rightarrow y_1(t) + y_2(t)$$

Thus, a linear system sends

$$a x_1(t) + b x_2(t) \rightarrow \boxed{S} \rightarrow a y_1(t) + b y_2(t)$$

for arbitrary a, b (which may be complex).

What space are we in?

Used to thinking in terms of mathematical functions:

$$\underset{\text{output}}{y} = f(\underset{\text{input}}{x})$$

For us, the system plays the role of the function, and signals are the input and output:

$$y(t) = S[x(t)]$$

This is why linearity looks different.

Example: linearity

Is the following system linear?

$$x(t) \rightarrow \boxed{S} \rightarrow x(t+1) - x(t-1)$$

$$x(t) \rightarrow y(t) = x(t+1) - x(t-1)$$

⇒ yes!

Test homogeneity

$$\begin{aligned} x'(t) &= ax(t) \Rightarrow y'(t) = x'(t+1) - x'(t-1) \\ &= ax(t+1) - ax(t-1) \\ &= a(x(t+1) - x(t-1)) \\ &= ay(t) \end{aligned}$$

Test additivity

$$\begin{aligned} x'(t) &= x_1(t) + x_2(t) \Rightarrow y'(t) = x'(t+1) - x'(t-1) \\ &= x_1(t+1) + x_2(t+1) - [x_1(t-1) + x_2(t-1)] \\ &= y_1(t) + y_2(t) \end{aligned}$$

$$x_1(t) \rightarrow y_1(t) = x_1(t+1) - x_1(t-1)$$

Exercise: linearity

Is the following system linear?

$$x[n] \rightarrow y[n] = x[n] + 1$$

Additivity **X**

$$x'[n] = x_1[n] + x_2[n]$$

$$\begin{aligned} y'[n] &= x'[n] + 1 \\ &= x_1[n] + x_2[n] + 1 \\ &\neq \underbrace{(x_1[n] + 1)}_{y_1[n]} + \underbrace{(x_2[n] + 1)}_{y_2[n]} \end{aligned}$$

Homogeneity **X**

$$x'[n] = a x[n]$$

$$\begin{aligned} y'[n] &= x'[n] + 1 \\ &= a x[n] + 1 \\ &\neq a y[n] \end{aligned}$$

Properties of systems: time invariance

A system is **time invariant** if a time-shifted input leads to an output time-shifted by the same amount.

$$x(t) \rightarrow y(t)$$

$$x(t - t_0) \rightarrow y(t - t_0)$$

Intuition: behaviour of the system is fixed over time.

Example: time invariance

Is this system time-invariant?

$$y(t) = \cos(3t)x(t)$$

$$x(t - t_0) = x'(t)$$

↓

$$\begin{aligned} y'(t) &= \cos(3t) \cdot x'(t) \\ &= \cos(3t) x(t - t_0) \end{aligned}$$

$$\neq y(t - t_0)$$

if time invariant
should get

$$y(t - t_0) = \cos(3(t - t_0))x(t - t_0)$$

≠

side note: memoryless?

Exercise: time invariance

Is this system time-invariant?

$$y(t) = x(t+1) - x(t-1) \quad \text{yes!}$$

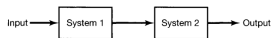
$$y(t-t_0) = x(t-t_0+1) - x(t-t_0-1)$$

Try $x'(t) = x(t-t_0) :$

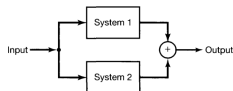
$$\begin{aligned} y'(t) &= x'(t+1) - x'(t-1) \\ &= x(t+1-t_0) - x(t-1-t_0) \\ &= y(t-t_0) \quad \checkmark \end{aligned}$$

LTI systems

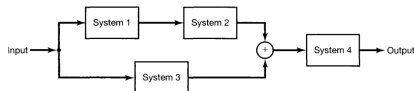
We are most interested in systems that are both **linear** and **time-invariant**, i.e., LTI systems.



(a)



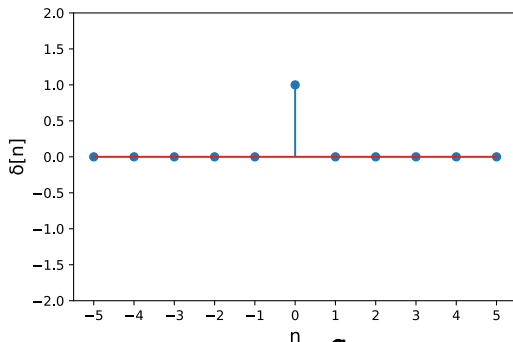
(b)



How can we characterize the behaviour of LTI systems?

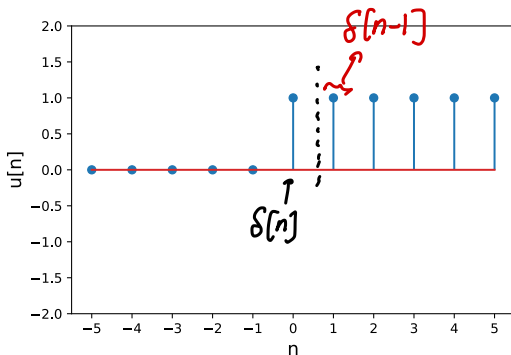
Image credits: Signals and Systems 2nd ed., Oppenheim

The DT unit impulse



$$\delta[n] = \begin{cases} 1 & n=0 \\ 0 & n \neq 0 \end{cases}$$

The DT unit step



$$u[n] = f(\delta[n])?$$

$$\delta[n] = f(u[n])?$$

$$u[n] = \begin{cases} 1 & n \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

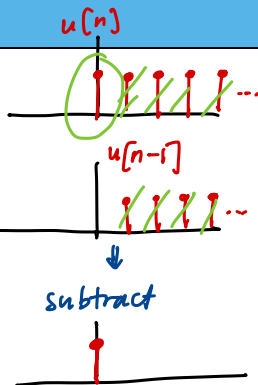
Relationships between basic signals

We can express these in terms of each other:

$$\delta[n] = u[n] - u[n-1]$$

$$u[n] = \sum_{m=-\infty}^n \delta[m]$$

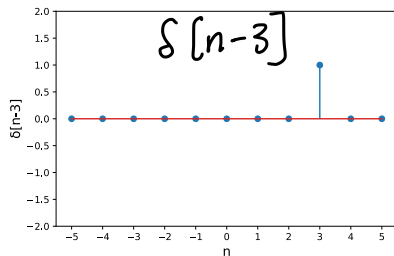
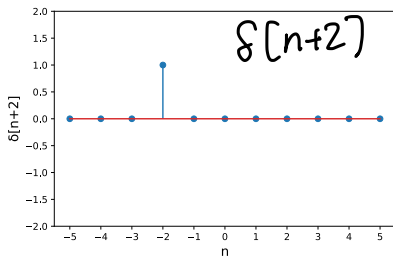
$$= \delta[n] + \delta[n-1] + \dots = \sum_{m=0}^{\infty} \delta[n-m]$$



The sifting property

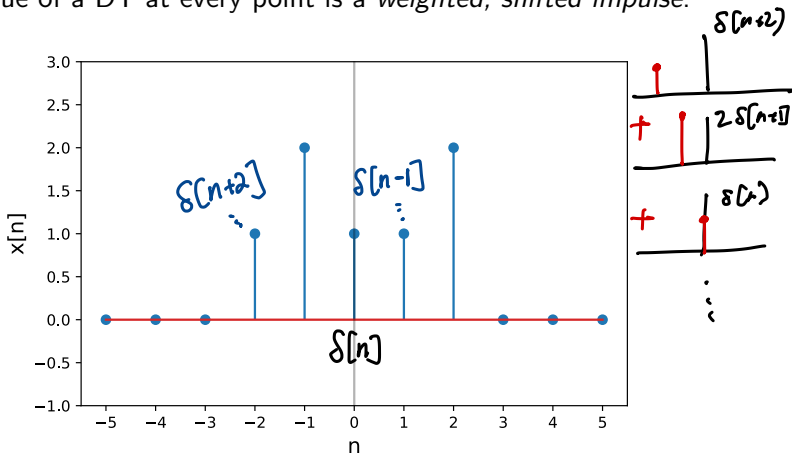
The unit impulse is an important tool for characterizing the behaviour of systems.

By considering unit impulses time-shifted as various points, we can pick out, or *sift* out specific parts of the signal.



The sifting property

The value of a DT at every point is a *weighted, shifted impulse*.



$$x[n] = \delta[n+2] + \delta[n] + \delta[n-1] + 2\delta[n-2] + 2\delta[n-1]$$

The unit impulse as a sampler

Multiplying by a shifted impulse “samples” the signal at that point:

$$x[n] \delta[n-k] = x[k]$$

Any signal can be written as a **superposition of weighted impulses**.

$$x[n] = \sum_{k=-\infty}^{\infty} x[k] \delta[n-k]$$

↑ ↑
Scale shifted impulse

For next time

- Content: • *Start w/ DT impulse response and convolution sum.*
- Continuous-time unit impulse and step
 - Convolution integral
 - Characterizing systems with the impulse response

Action items:

- ① Work on practice Assignment 1

} will post later in week

Recommended reading:

- from today's class: Oppenheim 1.6.5-6, 1.4, 2.1
- practice problems: 1.16-1.20, 2.1-2.7, 2.21
- for next class: Oppenheim 1.4, 2.2-2.3